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B. E. (Seventh Semester) Examination,
Oct.-Nov. 2015

(New Scheme)

(Computer Science & Engg.)

PARALLEL PROCESSOR & COMPUTING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all the questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.

a) Justify the use of Big O, Omega and Theta notations in analyzing algorithm complexity. 2

b) What is Cache Coherence and Cache coherence protocols? 7

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(c) Explain the hardware and software methods of implementing parallelism in uniprocessor systems.

(d) Explain 6 parallel algorithm models.

2. (a) A non-pipelined system can be converted into pipelined system, how?

(b) Given the following reservation table form the Collision Vector and find Minimum Average Latency.

	t0	t1	t2	t3
S1	X			X
S2		X		
S3			X	

(c) Draw and explain a floating point arithmetic pipeline for performing addition.

(d) Draw a functional block diagram of CRAY-1 and specify its hardware and software specifications.

3. (a) Explain VLIW computers.

(b) Find the 3rd and 5th smallest element using tree based selection algorithm from a set of data

8, 2, 9, 0, 1, 3, 5, 12, 11

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(c) Explain each step to perform shear sort in a 2D-Mesh having the following data : 7

1, 12, 21, 4, 15, 5, 22, 3, 9, 18, 14, 7, 17, 20, 13, 2

(d) Explain each step to perform Batcher's bitonic sort on a hypercube having the following data : 7

20, 1, 12, 18, 14, 7, 2, 15

4. (a) What is Emulation and how is it useful in developing parallel algorithms? 2

(b) Suppose we want to have a multiprocessor system that is scalable, then which kind of architecture do we need to choose. Also suggest a solution if there is cache coherence problem in your proposed system. 7

(c) "Multithreading is a practical mechanism for latency hiding", justify this statement. 7

(d) Explain any 6 capabilities of UNIX that makes it suited operating system for parallel processor architecture. 7

5. (a) Explain in brief : 2

(i) Parallelizing compilers

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(ii) Data-parallel programming

(iii) Shared-variable programming

(iv) Functional programming

(b) What is Interconnection networks for message passing and explain its three types.

(c) Draw and explain CC-NUMA Stanford Dash.

(d) Draw and explain functional view of an associative memory / processor. Also explain the four classes of AP (Associative Processors). 7

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